

High-Speed Digital Market and Technologies

The need of the hour is to utilise test and measurement products capable of mitigating the design and implementation challenges in deploying digital technologies



SADAF ARIF SIDDIQUI
TECHNICAL MARKETING SPECIALIST,
AGILENT TECHNOLOGIES INDIA

Worldwide, electronics applications are transforming towards digital capture, storage and transfer of communication. There are constant efforts to design and use high-speed interfaces to fulfil our communication needs.

In today's digital world, a single computer chipset integrates memory, peripheral component interconnect express (PCIe), serial advanced technology attachment (SATA), USB, HDMI and DisplayPort (DP) for connecting to devices and then fan out to additional technologies of DigRF, Ethernet, etc. Initially, digital interface USB 1.0 existed, which offered 1.5Mbps (low-speed) data transfer rates. But today, it has improved to USB 3.0, which takes only 54 seconds to transfer 16 GB of data, as against about six hours on a USB 1.0 port.

Memory data rates have increased over fifteen times, SATA by four times, PCI by more than three times and USB by ten times in a couple of years (see Table).

Interface Speeds

Technologies	Generation	Speed
PCI Express	Gen I	2.5 gigatransfers per second (GT/s)
	Gen II	5 GT/s
	Gen III	8 GT/s
SATA	Gen I	1.5 Gbps
	Gen II	3.0 Gbps
	Gen III	6.0 Gbps

Data Transfer Speeds With USB

	Song	USB Flash	USB Flash	HD-Movie
	4 MB	1 GB	16 GB	25 GB
USB 1.0	5.3 seconds	22 minutes	5.9 hours	9.3 hours
USB 2.0	0.1 second	33 seconds	8.9 minutes	13.9 minutes
USB 3.0	0.01 second	3.3 seconds	53.3 seconds	70 seconds

Another emerging digital link is mobile high-definition link (MHL). HDMI port, which is found on many LCDs, laptops and DVD players today, is not included in mobile phones due to their small size and form factor. Even with HDMI mini connectors, the mobile phone manufacturers do not want to put another connector on this very small device. MHL addresses the emerging model of streaming high-definition content from the mobile device to a television. In order to do this, a new single-lane interface was created that made use of the ubiquitous micro USB connector on every mobile device.

One aspect being widely observed in digital technologies is transition from parallel to serial. Examples in the world of computer design are SATA and PCIe. Similar trends are apparent in consumer and automotive electronics design too.

But what is causing the move to serial

data transmission? As parallel data rates reach the 1-2GHz range, clocking and skew challenges become very difficult to manage. Serial signaling with embedded clocking offers a way to achieve higher overall data transfer rates. However, these type of signals bring their own design challenges. The emergence of 'microwave digital' buses has pushed design and validation teams, as well as technology architects and standards groups, into unfamiliar territory. These digital signals are subjected to analogue parametric issues such as impedance mismatch, noise, jitter and cross-talks when these are implemented into real systems with PC board traces, connectors and cables.

At these high data rates, bit unit intervals become so small that system noise, cross-talk from adjacent electrical signals or even electromagnetic interference can impair the signal integrity. In addition, channel skin effects and a lossy signal path can further degrade the signal integrity, which can degrade the bit-error rate (BER) performance at the device receiver. These impairments can negatively impact the reliability and robustness of the digital input/output link performance, which is critical for any application or product.

The need of the hour is to utilise test and measurement products capable of mitigating the design and implementation challenges in deploying digital technologies, and market the products fast. Products such as high-bandwidth oscilloscopes, signal generators for receiver/stress tests, dedicated compliance suites for high-speed serial standards, logic analysers meant for physical layer testing and protocol analysers meant for protocol layer testing enable engineers to design, validate and develop products fast. ●